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DEPARTMENT OF COMPUTER SCIENCE ENGINEERING

MAJOR PROJECT

ACADEMIC YEAR : 2026-2027

CLASS : IV Year CSE-A

Title of project: Agrolens:AI-Driven crop Disease Detection System

ABSTRACT

Agriculture is essential for food production, but crop diseases can greatly affect yield and quality if not detected early. AgroLens: AI-Driven Crop Disease Detection System uses Artificial Intelligence and a Convolutional Neural Network (CNN) to identify crop diseases from leaf images with high accuracy. Through a user-friendly web application, farmers can upload leaf images and receive instant disease predictions along with treatment recommendations. The system enables early disease detection, reduces crop losses, improves agricultural productivity, and promotes sustainable farming practices.

It minimizes the need for manual inspection, saving time and effort for farmers, while providing a cost-effective solution for crop disease diagnosis. The proposed system offers an efficient, accurate, and scalable approach that supports modern smart agriculture and enhances overall farming productivity.

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